CHRIST (Deemed to be University)
Department of Computer Science
Master of Computer Applications

Course: MCA412-2 - Research Methodology

LAB Exercise III –

Defining Research Problem.

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Domain :

Marine Environmental Monitoring & Coastal Ecosystem-Preservation

Lab 3

Lab Exercise III:
Defining Research Problem.

1. What is Missing or Could Be Improved

An exhaustive synthesis of current marine litter monitoring reveals that satellite remote sensing and hydrodynamic modeling operate in complete isolation. Optical indices such as the Floating Debris Index (FDI) fail in sediment-laden estuarine zones because suspended particulate matter (SPM) mimics sub-pixel debris reflectance. Furthermore, existing classifiers cannot reliably differentiate weathered plastics from natural organic flotsam (driftwood and macroalgae) using spectral reflectance alone. Crucially, current Earth observation workflows treat plastic detections as static snapshots, entirely omitting hydrodynamic current vectors and windage factors that govern where nearshore debris actually aggregates and beaches.

2. New Questions That Arise

- Can an adaptive, sediment-compensated spectral baseline mathematically isolate polymer reflectance from high suspended sediment backscatter in coastal estuaries?
- To what extent can the fusion of Gray-Level Co-occurrence Matrix (GLCM) texture metrics with SWIR band ratios resolve the spectral ambiguity between plastic debris and natural organic materials?
- How accurately can localized Eulerian-Lagrangian velocity vectors predict the immediate coastal trapping and accumulation zones of satellite-detected debris fields?
- How does atmospheric correction uncertainty in coastal transition waters propagate into false-positive debris identifications?
- Can cross-sensor validation using multi-temporal SAR coherence identify macroplastic aggregations when optical satellite data is obscured by cloud cover?

3. Research Questions

- RQ1: How can an adaptive baseline index decouple suspended particulate matter (SPM) backscatter from sub-pixel polymer reflectance to enable reliable plastic detection in turbid estuarine waters using Sentinel-2 MSI data?
- RQ2: To what degree does the integration of multi-spectral GLCM texture features improve the classification separability of marine debris from look-alike natural organic flotsam across heterogeneous coastal waters?
- RQ3: What is the empirical correlation between nearshore hydrodynamic convergence vectors (derived from Copernicus reanalysis) and multi-temporal satellite debris hotspots along estuarine boundaries?

4. How My Research Could Contribute to Filling These Gaps

My research will formulate an end-to-end computational methodology that bridges optical remote sensing and hydrodynamic transport physics. By utilizing the MARIDA benchmark and raw Sentinel-2 MSI scenes, I will formulate a turbidity-compensated spectral indexing algorithm coupled with GLCM texture metrics to suppress false positives caused by sediments and natural flotsam. Furthermore, I will spatially synchronize these satellite detections with Copernicus Marine Service (CMEMS) hydrodynamic current velocity vectors to compute nearshore convergence indices, providing the first unified exploratory framework for tracking nearshore debris trajectory and beaching risk.

5. Future Research Directions Suggested by the Authors

- Biermann et al. advocate developing robust algorithmic adaptations capable of maintaining plastic detection performance in turbid, optically complex coastal and riverine environments.
- Kikaki et al. (MARIDA) recommend combining spatial texture analysis across multiple window sizes with advanced feature selection to distinguish polymers from natural organic matter.
- van Sebille et al. call for the explicit integration of physical hydrodynamic transport frameworks with Earth observation data to reconcile surface detections with sub-surface advection.
- Guffogg et al. highlight the need for empirical spectral libraries of mixed, weathered coastal polymers to evaluate sub-pixel downscaling algorithms.

6. Two Futuristic Directions

- Physics-Informed Neural Operator (PINO) Coastal Drift Networks: Developing neural operators that ingest multi-spectral satellite imagery and directly solve continuous shallow-water hydrodynamic advection equations to forecast real-time 3D debris dispersion without requiring manual numerical grid setups.
- Autonomous Satellite-to-Swarm Telemetry Pipelines: Designing orbital edge-computing systems that detect floating debris plumes from satellite imagery in real-time and autonomously dispatch autonomous surface vehicles (ASVs) to localized hydrodynamic trapping zones for interceptive recovery.

7. Problem Statement

To resolve the critical failure of optical satellite indices in sediment-heavy estuarine waters, eliminate severe spectral confusion between polymers and natural flotsam, and overcome the absence of physical transport coupling in Earth observation workflows, this research aims to formulate a turbidity-compensated, texture-enhanced spectral detection framework using Sentinel-2 MSI data and empirically correlate satellite-derived debris hotspots with Copernicus nearshore hydrodynamic velocity fields to delineate coastal plastic convergence zones.